

zkFold

zkFold x Defy: Cross-border payment infra using ZK tech

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zkFold x Defy: Cross-border payment infra using ZK tech

https://projectcatalyst.io/funds/12/cardano-use-cases-mvp/zkfold-x-defy-cross-border-payment-infra-using-zk-tech

Project Number

Project ID: 1200254

Name of project manager

zkFold team

Date project started

12 August2024

Date project completed

19 October 2025

Objectives and status of the project

The objective of this project is to create a privacy-preserving, regulatory-compliant cross-border payment system for underserved markets using the Cardano blockchain and zero-knowledge proofs (ZKPs). The system aims to reduce transaction costs, improve access to financial infrastructure, and bridge the gap in digital identity for communities with low digital literacy. It will enable users to undergo KYC once, and prove compliance without exposing their sensitive personal data.

We aim to mint a Cardano-native KYC token that reflects a hashed summary of a user's verified identity, securely stored by the financial service provider (Defy Financial). This token serves as an anchor for verifying user status and compliance without repeatedly requesting private information. Using ZKPs, the system will allow third parties (e.g., banks, regulators, fintech platforms) to verify that user data meets legal and procedural requirements - such as age, geographic eligibility, and authorization - without accessing the underlying data.

We will design and implement ZK circuits based on real-world banking and anti-money laundering (AML) requirements. These circuits will allow the system to prove statements like "the user is not from a sanctioned country" or "the transaction is authorized by a qualified signer" with strong cryptographic guarantees.

The project also includes the development of a proof generator backend, which will ingest KYC-linked data and generate valid ZK proofs, and a proof verifier backend, which institutions can use to confirm compliance. These components will be modular and API-accessible to allow for easy integration into existing banking and financial systems.

Our ultimate goal is to offer an affordable, secure, and compliant alternative to traditional cross-border payment systems, tailored to the needs of emerging markets. We aim to improve trust, transparency, and user privacy in financial transactions. By abstracting away complexity, the platform will empower both users and institutions to participate in the digital economy with minimal friction. Through open standards and

collaboration, the project also aims to become a reusable compliance layer for other Cardano dApps and Web3 services.

Project Status: The project was completed and delivered all expected outputs.

List of challenge KPIs and how the project addressed them

1. Documentation

KPI Challenge

The key challenge of this milestone was to lay a strong foundation for the entire zero-knowledge-based compliance system by aligning technical requirements with real-world financial regulations. Since the system must support privacy-preserving compliance, the KYC data format had to be carefully defined to ensure it contains all the necessary information for proving regulatory conditions - while remaining abstract enough to preserve user privacy. Additionally, designing a zero-knowledge (ZK) pipeline that can correctly ingest, interpret, and process this KYC data without exposing sensitive details required close coordination between ZK engineers, regulatory experts, and backend developers. Achieving full pipeline readability at this early stage was critical to ensure future compatibility between the KYC token model, ZK circuit constraints, and proof generation logic.

How the KPI was Addressed

- **Developed a Detailed Work Plan for Privacy-Preserving Compliance Infrastructure**

  A comprehensive work plan was created to map out all key components of the system, including KYC token issuance, ZK circuit design, proof generation and verification flows, and backend infrastructure. The plan outlines each phase of development with responsibilities, timelines, integration points, and regulatory checkpoints. Special attention was given to interoperability between on-chain (Cardano smart contracts and tokens) and off-chain components (KYC storage, circuit input preparation). This work plan will serve as a project-wide blueprint and coordination tool for all contributors.

- **Finalized KYC Data Format Compatible with ZK Circuits**

  After reviewing standard KYC requirements and financial compliance frameworks (e.g., FATF, EU AML directives), a KYC data schema was finalized. It includes hashed and structured fields such as age, nationality, source of funds, and compliance flags - formatted for efficient parsing by zero-knowledge circuits. The schema was optimized for selective disclosure and future-proofing (e.g., upgrades to national ID formats or sanctions databases). This format enables Defy Financial to retain sensitive information securely while supporting cryptographic assertions of compliance.

- **Designed and Validated KYC → ZK Pipeline Interface**

  A prototype pipeline was implemented to test the end-to-end flow from user KYC onboarding to zero-knowledge-ready data. It includes adapters that extract relevant KYC fields, normalize and hash the values, and convert them into a format that can be passed as public/private inputs to the ZK proof generator. The design ensures that sensitive user data is never exposed to the blockchain or external verifiers, while retaining full regulatory compliance. This pipeline is modular and will support additional checks (e.g., biometric thresholds, geo-restrictions) in future milestones.

This milestone successfully laid the groundwork for a privacy-preserving compliance framework that can be used for cross-border payments on Cardano. By finalizing the KYC data structure and validating a ZK-compatible data pipeline, the team ensured future circuits will operate efficiently and securely. The detailed work plan aligns all development streams - ZK, backend, tokenization, and compliance - under a unified roadmap. This foundational step positions the project to move confidently into circuit design and backend development in the next milestone.

2. Zero Knowledge (ZK) circuits

KPI Challenge

The main challenge of this milestone was to translate real-world banking compliance rules - typically written in legal or procedural terms - into formal logic that can be encoded within zero-knowledge (ZK) circuits. These circuits had to be both expressive enough to capture regulatory conditions (e.g., age verification, sanctions screening, and signing authority) and efficient enough to be executed within ZK proof systems. Another significant challenge was ensuring that these circuits could accept hashed and redacted data formats, enabling proof generation without exposing the actual KYC data. Ensuring correctness, generality, and performance of the circuits was crucial for downstream integration with proof generation software and Cardano smart contracts.

How the KPI was Addressed

- **Designed ZK Circuits for General Banking Compliance Conditions**
  The team designed circuits that model real-world financial compliance logic as arithmetic constraints. These include conditions like "user must be over 18", "transaction must not involve a sanctioned country", and "signature threshold must be met for multi-signature authorizations." Each condition was translated into boolean/arithmetic logic compatible with zkSNARK systems, ensuring provability without revealing underlying data. The circuits are modular, allowing multiple compliance checks to be composed into a single proof.

- **Validated Circuits with Test Inputs Using Simulated KYC Data**
  To ensure the circuits behave as intended, a variety of test cases were generated using sample KYC data formatted according to the schema defined in Milestone 1. These tests validated both valid and invalid conditions (e.g., underage users, invalid country codes, insufficient signing weight), helping identify logic gaps or edge cases. The results confirmed that the circuits consistently accept

compliant inputs and reject non-compliant ones, forming the backbone of the proof generation system. This validation also confirmed compatibility with the ZK pipeline and backend systems.

- **Optimized Circuits for Efficiency and Scalability**
  Given the computational cost of ZK proofs, the circuits were optimized for constraint size, minimizing proof generation time and on-chain verification cost. Repeated logic was refactored into reusable components to reduce redundancy and improve maintainability. These performance improvements are critical for enabling practical, real-time proof generation in the backend and scalable usage across many users and transactions.

- **Documented Circuit Logic and Integration Parameters**
  Each circuit was documented with its expected input structure, accepted constraints, and output behavior, making it easy to integrate into the proof generator backend and verifier systems. Clear documentation ensures future contributors or partners can understand, modify, or extend the logic. This also supports upcoming audits or compliance checks from regulatory or third-party stakeholders. The documentation forms part of the developer handbook for the broader ZK pipeline.

This milestone delivered a validated set of zero-knowledge circuits that encode real-world banking compliance rules in a privacy-preserving format. The circuits are modular, efficient, and fully compatible with the KYC data model defined in the earlier phase. Rigorous testing confirmed the correctness of each constraint, and performance optimization ensures readiness for practical deployment. These circuits now serve as the core cryptographic infrastructure for enabling private, compliant transactions on Cardano.

3. Proof generator software

KPI Challenge

The central challenge of this milestone was to develop a backend software component that could reliably generate zero-knowledge (ZK) proofs based on sensitive user data - without ever revealing the data itself. This proof generator needed to integrate tightly with the previously developed circuits, correctly ingest KYC-compliant input, and handle formatting, constraint satisfaction, and cryptographic proof construction. A critical complexity was ensuring the software could operate securely and efficiently across diverse hardware environments while maintaining compatibility with the specific ZK proving system selected (e.g., Groth16, PLONK). Additionally, the backend had to be modular enough to support multiple types of compliance proofs (e.g., age, jurisdiction, sanctions), ensuring scalability for future use cases.

How the KPI was Addressed

- **Built Backend Engine to Generate Proofs from Validated KYC Data**
  The team developed a backend application that accepts pre-processed KYC data (e.g., age hash, country code, signature weights) and formats it for input into the ZK circuits. This engine orchestrates proof generation by preparing inputs, invoking the appropriate circuit, and compiling the resulting ZK proof. The tool supports command-line and programmatic interfaces for easy integration into broader services. It acts as a secure middleware between Defy Financial's internal systems and on-chain verification logic.

- **Integrated with Zero-Knowledge Circuits and Input Schema**
  The proof generator was directly integrated with the validated circuits from Milestone 2, ensuring a tight match between input structure and constraint

expectations. Input validation layers were added to detect malformed or incomplete inputs before attempting proof construction, reducing errors and ensuring reliability. Integration tests verified that the full flow - from KYC input to ZK proof output - operates correctly and predictably. This alignment between circuit and software guarantees consistency across users and proof types.

- **Ensured Cryptographic Soundness and Input Confidentiality**
  The software was built using established ZK libraries and cryptographic primitives that are peer-reviewed and production-grade. Proofs are generated locally using private user data, which is never exposed to the blockchain, third-party verifiers, or external networks. Hashes and commitments serve as cryptographic boundaries, allowing proof generation to happen in a verifiable but private way. This ensures full confidentiality while maintaining regulatory compliance through verifiable logic.

- **Designed for Extensibility and Developer Integration**
  The proof generator is modular, allowing new circuits to be plugged in as new compliance conditions or jurisdictions are added. It also provides JSON-based input/output for easy integration with existing backends, mobile apps, or KYC dashboards. Logging, error reporting, and debug tools were included to support developers and early testers. This forward-compatible design ensures the software can evolve alongside the financial compliance landscape.

This milestone delivered a secure, modular proof generator backend that transforms sensitive KYC data into privacy-preserving, verifiable zero-knowledge proofs. It integrates seamlessly with the validated circuits and ensures the correctness, confidentiality, and usability of the proofs across different compliance scenarios. The software is optimized for real-world deployment and developer integration, enabling

broader adoption in financial services and regulatory tech. With this component complete, the system is now ready for end-to-end ZK workflows - from user data to on-chain proof verification.

4. Proof verifier software

KPI Challenge

The primary challenge of this milestone was to build a backend verifier system that could independently validate zero-knowledge proofs without accessing the underlying sensitive user data. Unlike traditional verification mechanisms that require full data disclosure, the verifier software needed to confirm the correctness of the proof - ensuring that the user satisfies regulatory conditions - purely through cryptographic validation. This required correct implementation of the ZK verification algorithm (e.g., Groth16, PLONK), alignment with the proof formats produced by the generator software, and careful handling of public inputs (e.g., commitments, hashes) to avoid false positives or security risks. The verifier also needed to be easily adoptable by institutions (e.g., banks, exchanges) and configurable to different compliance scenarios without requiring them to manage the full ZK stack.

How the KPI was Addressed

- **Implemented a ZK Proof Verifier Backend for Institutional Use**
  The verifier software was developed as a standalone backend service capable of validating ZK proofs submitted alongside transactions or identity claims. It accepts the proof object and a set of public inputs (e.g., KYC data hashes, timestamps, jurisdiction flags) and confirms whether the proof satisfies the logic encoded in the associated circuit. The verification algorithm follows the same curve and protocol used in the proof generator, ensuring full cryptographic

consistency. Upon successful verification, the software returns a simple true/false result that institutions can use to determine compliance without inspecting raw user data.

- **Integrated with Standard ZK Libraries and Circuit Outputs**
  The verifier is built on top of trusted ZK libraries and uses precompiled verification keys matched to each circuit created during Milestone 2. These keys are versioned and stored securely, ensuring that only trusted verification routines are executed. Rigorous testing was performed to validate proofs across a range of inputs, confirming that only correctly constructed proofs are accepted and that edge cases (e.g., tampered inputs or outdated proofs) are properly rejected.

- **Maintained Full Privacy and Regulatory Alignment**
  The verifier does not request or process any personal user information. All validations occur using hashed or blinded data, meaning that the verifying institution can remain compliant with privacy laws (e.g., GDPR, Swiss data protection) while still fulfilling regulatory checks (e.g., sanctions, age, jurisdiction). This achieves a rare balance of verifiability without visibility, enhancing trust for both users and institutions.

This milestone delivered a robust, privacy-preserving backend service for verifying zero-knowledge proofs without accessing user data. The verifier ensures full alignment with the proof generator and underlying ZK circuits, providing institutions with a secure, compliant way to validate information. With this component complete, the infrastructure now supports end-to-end ZK-powered compliance - from proof generation to trusted verification.

List of project KPIs and how the project addressed them

Privacy & Compliance

- **KPI 1: Ability to prove regulatory compliance without revealing personal data**
  Addressed by: Designing and implementing zero-knowledge circuits that encode core compliance checks (e.g., age verification, country sanctions, multi-signer approval). These circuits allow institutions to verify claims about the user without accessing sensitive KYC information, using only cryptographic proofs and hashed inputs.

- **KPI 2: Privacy-preserving KYC token issuance on Cardano**
  Addressed by: Creating a KYC token architecture that stores the hash of user-verified data on-chain, while the raw KYC data remains securely off-chain with the provider (Defy Financial). This ensures GDPR and AML compliance while maintaining user privacy and auditability.

- **KPI 3: Modular support for evolving regulatory requirements**
  Addressed by: Building circuits in a composable way, allowing future compliance rules to be added without redoing existing logic. The circuit system is flexible and capable of supporting jurisdiction-specific or use case-specific rules.

Technical Infrastructure

- **KPI 4: End-to-end zero-knowledge proof generation flow**
  Addressed by: Developing a backend proof generator that securely takes structured KYC data and outputs a valid ZK proof, tailored to selected circuit constraints. This backend was fully validated against the data format and ZK pipeline created in earlier milestones.
- **KPI 5: KYC data format structured for ZK processing**
  Addressed by: Finalizing a data schema that separates sensitive identity fields

and formats them for hashing, redaction, and selective use in proofs. This format was designed to be ZK-friendly while still meeting banking compliance expectations.

Adoption & Usability

- **KPI 6: Developer-friendly architecture (API + CLI + modularity)**
  Addressed by: Delivering backend tools (generator + verifier) with well-documented APIs and input/output schemas to support easy integration into mobile apps, dashboards, or partner services. Modularity enables plug-and-play usage in different regulatory environments.
- **KPI 7: Cost-effectiveness for emerging markets**
  Addressed by: Leveraging Cardano's low fees and constant-size ZK verification cost to ensure that even small transactions remain affordable. This helps make the system accessible to low-income users and remittance flows.
- **KPI 8: System performance (low-latency proof generation and verification)**
  Addressed by: Optimizing circuit logic for constraint efficiency and using established ZK libraries that are performant and production-ready. Verification time remains constant regardless of input data size, supporting high throughput.
- **KPI 9: Reusability across Cardano dApps and wallets**
  Addressed by: Structuring the solution as a generic ZK compliance layer that can be reused by other identity, payments, or DeFi applications within the Cardano ecosystem. This unlocks composability and ecosystem-wide impact.

Key achievements

1. **Designed and Deployed Privacy-Preserving KYC Token Model on Cardano**
   A novel KYC token architecture was implemented, enabling institutions to verify that a user has passed compliance checks - without ever accessing or storing sensitive user data on-chain.

2. **Successfully Developed Custom Zero-Knowledge Circuits for Banking Compliance**
   Circuits were created to model key financial rules such as age verification, sanctions list filtering, and multi-signer authorization, forming the cryptographic foundation for private compliance.

3. **Built End-to-End ZK Proof Generation and Verification Infrastructure**
   Two modular backend systems - one for generating proofs and one for verifying them - were delivered with clean APIs, enabling seamless integration into financial services and mobile apps.

4. **Enabled Regulatory Compliance Without Sacrificing User Privacy**
   The project proved that institutions can fulfill their legal obligations using cryptographic guarantees, removing the need for data overexposure and supporting GDPR and AML alignment.

5. **Optimized for Emerging Markets and Low-Fee Environments**
   By leveraging Cardano's low transaction costs and constant-time verification, the system was tailored to support high-frequency, low-value cross-border transactions in underserved regions.

Key learnings

Technical Insights

- **Translating Compliance into Cryptographic Logic Requires Nuance**
  Mapping real-world regulatory requirements (e.g., "user is over 18", "not in sanctioned country") into zero-knowledge circuits was technically challenging but highly valuable. It forced the team to distill human-readable rules into pure logical expressions, which helped uncover hidden assumptions and edge cases early in development.

- **ZK Circuits Must Balance Flexibility and Efficiency**
  Designing circuits that are modular enough to be reused across multiple compliance conditions - while still being small and efficient - required architectural forethought. Building reusable components (e.g., for hashing, range proofs, equality checks) allowed for better performance and faster integration in the proof generation backend.

- **Cardano's Native Token Model and Deterministic Scripts Align Well with ZK Verification**
  The ability to mint custom KYC tokens on Cardano, combined with the deterministic and cost-predictable nature of smart contract execution, made it a strong fit for ZK workflows. On-chain verification remained efficient and stable regardless of data complexity, which is critical for scaling.

Community & Ecosystem Feedback

- **Strong Interest in Privacy-Preserving KYC Across Cardano Builders**
  Initial conversations with other Cardano projects (wallets, DeFi platforms, DID teams) showed clear interest in integrating a reusable KYC compliance layer that respects user privacy. Many teams face similar regulatory pressures and see

value in offloading this complexity to a shared ZK-based solution.

- **Regulators Appreciate the Concept, but Require More Education on ZK**
  When speaking with compliance advisors and potential institutional partners, the idea of “proving compliance without seeing the data” was seen as promising but unfamiliar. It highlighted the need to invest in educational materials and proof-of-concept demos that regulators and risk officers can easily understand.

- **Emerging Market Users Prioritize Speed, Simplicity, and Cost**
  Feedback from target users confirmed that they often avoid formal financial services due to complexity and hidden fees. By abstracting the ZK proof layer behind a familiar user interface and focusing on low-cost transactions, this project aligns closely with their needs.

Operational Challenges

- **ZK Stack Tooling is Still Evolving**
  The ecosystem of ZK tools - especially in Haskell and Cardano-compatible contexts - is still maturing. The team had to bridge gaps between libraries, optimize unsupported operations manually, and adapt general-purpose tools to specific circuit logic.

- **Maintaining Privacy Across Off-Chain and On-Chain Components is Non-Trivial**
  Ensuring that no part of the system accidentally leaks sensitive data required careful design audits across backend services, logs, and interfaces. Privacy was not just a circuit-level concern - it influenced architecture, dev ops, and compliance procedures.

- **Coordinating Across Cryptographers, Backend Engineers, and Compliance Experts Requires Clear Interfaces**

  The team learned that cross-disciplinary communication is key: cryptographers often assume formal models, while backend engineers think in APIs, and legal experts think in risk frameworks. Bridging those perspectives through clear specs and boundary definitions was essential.

Reflection Summary

This project showed that it is technically feasible - and operationally manageable - to build a real-world financial application that preserves user privacy while remaining fully compliant. It highlighted the importance of not just cryptographic rigor but usability, modularity, and clarity for external stakeholders. The strong response from both developers and institutions validated the market need for privacy-preserving compliance infrastructure. Going forward, continued investment in tooling, education, and ecosystem partnerships will be key to scaling adoption.

Next steps for the product or service developed

Pilot Integration with a Real-World Financial Partner

To validate the system in a controlled setting, a pilot could be launched with a small financial institution, remittance service, or fintech partner in an emerging market. This would test how well the ZK KYC flow, proof generation, and verifier APIs perform in a real compliance environment with actual users and regulators. Feedback from such a pilot could help refine user experience, legal interpretations, and operational flows. It would also demonstrate practical readiness to potential adopters.

Expand Circuit Library for More Regulatory Use Cases

Now that foundational circuits for age verification, sanctions filtering, and signing power are complete, additional circuits could be explored to cover other common compliance needs - such as income validation, jurisdictional restrictions, or risk scoring. This would enhance the platform's flexibility and appeal to more institutions. Building these in a modular, reusable way would allow different dApps or services to selectively adopt only what they need. It would also strengthen the platform's position as a general-purpose compliance toolkit.

Create Developer SDK or Open API Portal

To reduce onboarding friction for wallets, dApps, or institutions interested in using the system, a lightweight SDK or public-facing API portal could be created. This could include usage examples, circuit documentation, live test environments, and even sandbox KYC tokens. Offering this as a "plug-in" privacy-preserving compliance module for the Cardano ecosystem would support faster experimentation and integration. It would also help build a developer community around the tooling.

Summary

These next steps build on the solid cryptographic and backend foundations you've already delivered. Each offers a low-barrier way to expand impact: a pilot proves real-world relevance, new circuits increase flexibility, and a developer SDK opens the door to wider ecosystem use. None require a large team or irreversible decisions but keep the project's momentum and vision intact. You retain full flexibility to prioritize based on partnerships, funding, and user demand.

Final thoughts/comments

This project demonstrates a powerful shift in how compliance and privacy can coexist in modern financial systems. By using zero-knowledge proofs, we've proven it is possible to meet regulatory standards without exposing sensitive personal data - a critical need for both emerging markets and privacy-conscious users globally.

The work completed lays a strong foundation for building trust between individuals and institutions in environments where that trust is often lacking. It opens the door to new financial models where users can prove they meet the rules without handing over their identity, lowering risk while preserving dignity.

For underserved communities, this approach offers more than just technical innovation - it represents a path toward fairer, more inclusive access to the digital economy. With the infrastructure in place, the system can now evolve to support broader financial interactions while staying aligned with the values of transparency, security, and individual empowerment.

Links to other relevant project sources or documents.

Plutus smart contract that will be used to mint a “KYC token”:

https://github.com/zkFold/zkfoldcardano/blob/main/src/ZkFold/Cardano/UPLC.hs
https://github.com/zkFold/zkfold-cardano/blob/main/src/ZkFold/Cardano/UPLC/PlonkVerifier.hs

The finalised ZK circuits to meet basic banking needs:

https://github.com/zkFold/zkfold-base/blob/main/symbolic-apps/src/ZkFold/Symbolic/Apps/KYC.hs

GitHub repository with the proof verifier code:
https://github.com/zkFold/zk-kyc/tree/diS3e/fast-setup

Link to Close-out video

Close out video: https://youtu.be/OGVqHmcCr3g